

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I G. Vignesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

G. Vignesh

Signature of the student:

2. Scenario: Intercepted Suspicious Message.

Given Ciphertext:

GSRH RH Z HVXIVG.

→ This uses the Atbash cipher, where the alphabet is reversed:

* A ↔ Z O ↔ L
 B ↔ Y P ↔ K
 C ↔ X Q ↔ J
 D ↔ W R ↔ I
 E ↔ V S ↔ H
 F ↔ U T ↔ G
 G ↔ T U ↔ F
 H ↔ S V ↔ E
 I ↔ R W ↔ D
 J ↔ Q X ↔ C
 K ↔ P Y ↔ B
 L ↔ O Z ↔ A

 M ↔ N

Therefore:

GSRH RH Z HVXIVG

THIS IS A SECRET

i) Original message:

"This is a Secret."

ii) Justification:

This message has the pattern.

* The word lengths are 4-2-1-6, which match exactly.

Also, the repeated word structure is preserved.

* RH → appears as a two-letter word - IS

* Z → appears as a one-letter word - A

* HVXIVG has six letters → SECRET

⇒ The letter substitutions are consistent with Atbash,

a simple, substitution cipher in which the alphabet

is reversed.

* Original message ⇒ This is a Secret.

5. Scenario: Legacy Encryption in Corporate Communication.

* Ciphertext:

WECLTEERDSOEFEAEACIVDEN.

1. Original message:

This ciphertext is produced by a transposition cipher, most commonly identified as a 3-level/3-column rail-style transposition.

→ Decryption.

Reversing the three-level arrangement gives:

WEAREDISCOVEREDFLEEATONCE

Adding Spaces:

We are discovered flee at Once.

2. Why it is vulnerable.

→ Letters are not changed, only rearranged.

→ Different arrangements can be tested easily.

→ It provides weak confidentiality.

3. Enhancement.

Replace the transposition cipher with -

AES-256-GCM

Plaintext $\rightarrow$ AES-256-GCM $\rightarrow$ Ciphertext.

* This provides strong confidentiality and integrity for modern communication.